Occupational Gender Bias in Ungendered Languages and LLMs: Comparing Hungarian and Chinese

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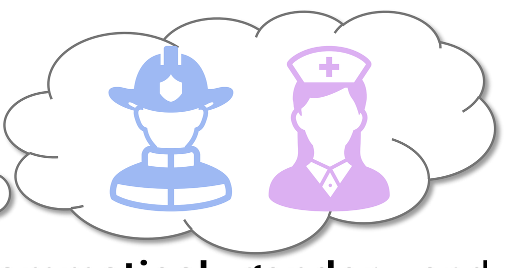
Introduction

Occupational gender stereotypes: generalizations about the roles and characteristics of individuals in a certain group of the workforce.



- People often **associate** certain professions with **men** or **women**.
- Some languages have **grammatical gender**, and occupational nouns are **explicitly gendered**.

actor		actress
acteur	FR; NL	actrice
attore	IT	attrice
Schauspieler	DE	Schauspielerin
अभिनेता *abhinetā*	HI	अभिनेत्री *abhinetrī*

- Lot of research on Indo-European languages with grammatical gender, **less** on other language families.
- How about occupational gender **bias** in **linguistically ungendered languages**?

Hungarian & **Chinese** | 匈牙利语和中文
- No grammatical gender
- Use mostly gender-neutral occupational nouns

Human vs. **AI** | 人类与人工智能
- LLMs train on huge amounts of human-generated text;
- If **occupational gender stereotypes** are **embedded** in language, the models will **reproduce** them.

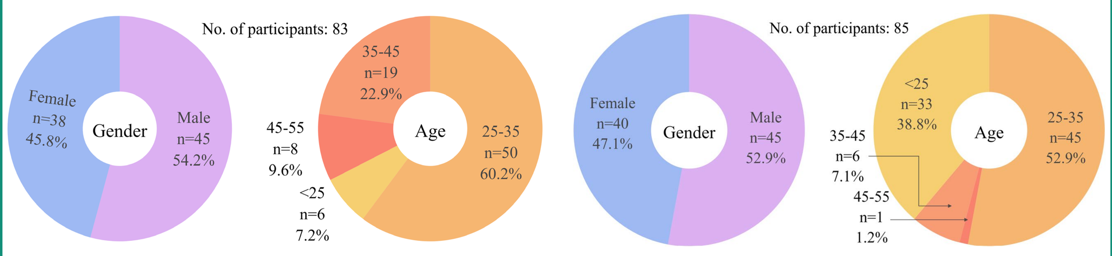
How do speakers of ungendered languages deal with gender stereotypes on the lexical level?

Research Questions:
- To what extent do speakers of Hungarian and Chinese exhibit **gender bias** when rating job titles?
- Are there differences between **male** and **female** raters?
- What are the key **differences** and **similarities** in occupational gender stereotypes between Hungarian and Chinese speakers?
- How **similar** are human and LLM judgments?

Experiments

Study 1: Human Ratings

- **Survey-based ratings**; 7-point Likert scale, from *completely male* (-3) to *completely female* (+3); Participants were asked to rate on *"how likely an occupation is to be pursued by men or by women."*
- **Participants**: HU N=83 (38 female) | ZH N=85 (40 f)



- **Materials**: 44 linguistically gender neutral occupational human agent nouns and job titles, e.g.:

Hungarian	Chinese	English
fodrász	理发师	hairdresser
programozó	程序员	programmer
orvos	医生	doctor
…	…	…

+ 6 attention checks

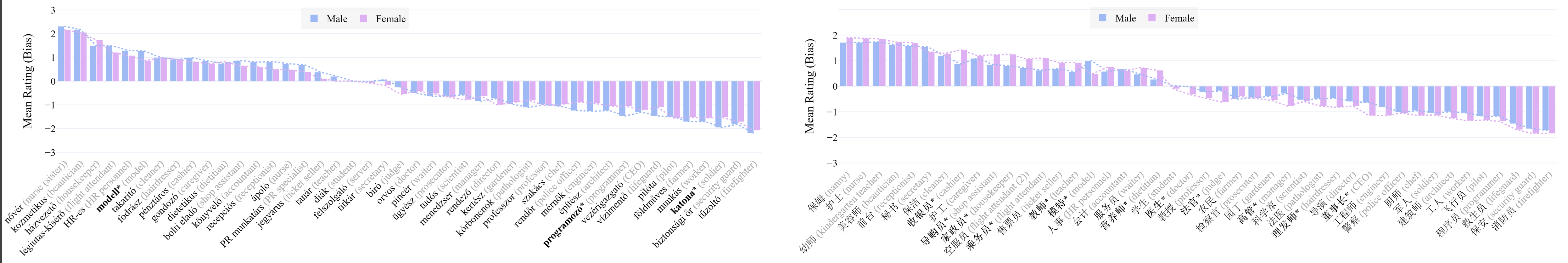
- **Procedure**: Online via Microsoft Forms

- **Methods**: t-tests (`scipy`) and visualizations (`plotly`) in Python; Cumulative Link Mixed Models (`clmm`) in R: Rating ~ Language * Gender + (1 | Participant) + (1 | Item); Spearman's ρ to compare human with LLM ratings.

Spearman's rank correlation coefficient

Results & Findings

1. Occupational stereotypes emerged in **both languages**, according to expectations.
2. Participant gender may **influence** the intensity of occupational gender stereotypes:



3. Most stereotypes were shared **cross-linguistically**, but there were differences:

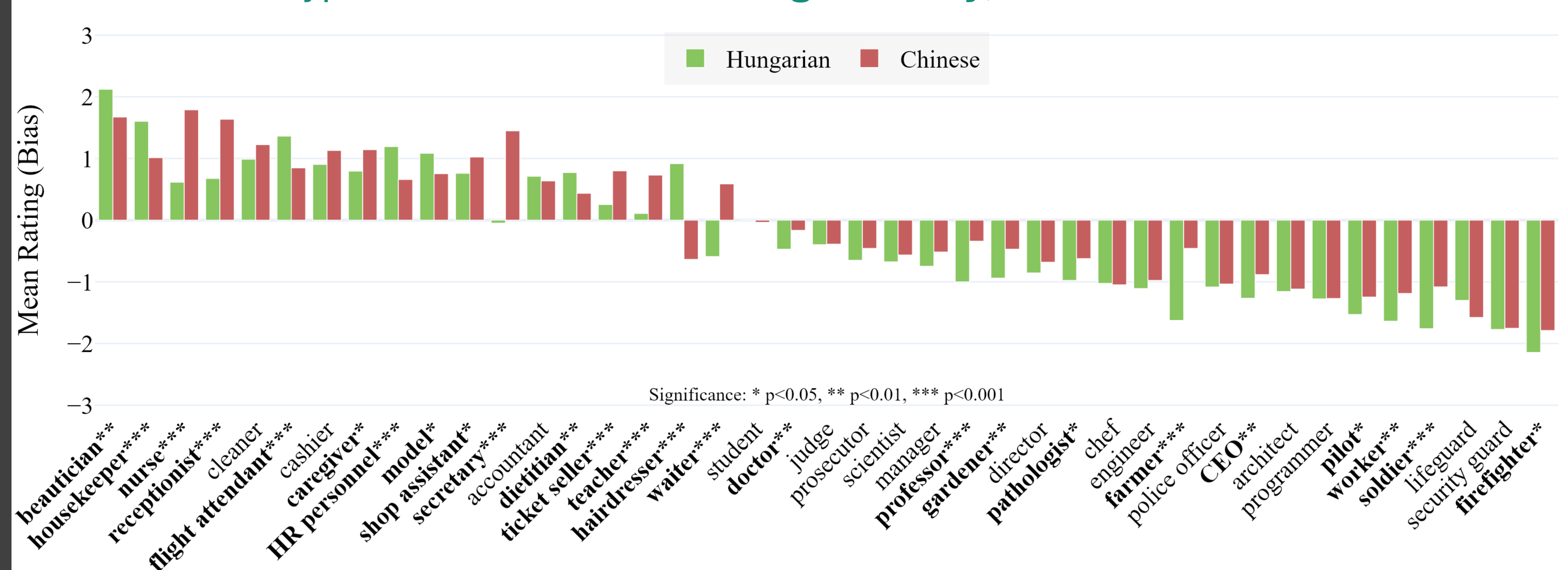
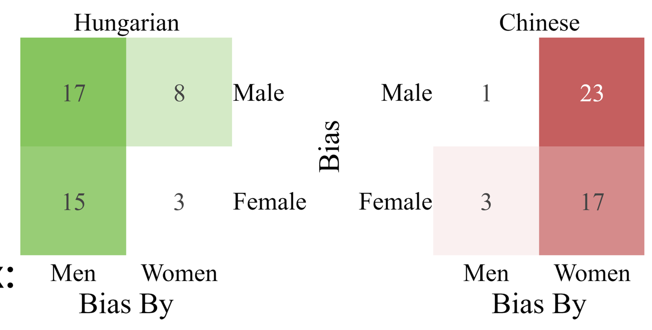


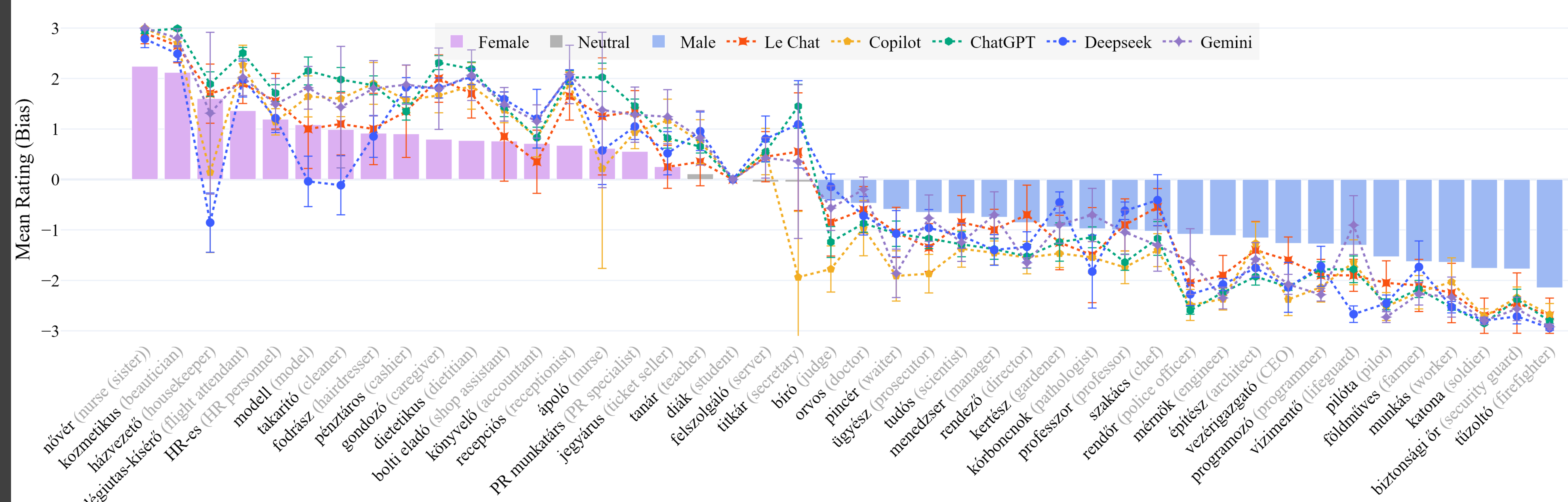
Table 2: Top 10 most diverging occupations between Hungarian and Chinese participants, sorted by the absolute difference between the mean ratings.

Occupation	HU	ZH	Diff.	p
hairdresser	0.92	-0.64	1.55	<.001
secretary	-0.05	1.45	1.50	<.001
waiter	-0.59	0.59	1.18	<.001
farmer	-1.63	-0.46	1.17	<.001
nurse	0.61	1.79	1.17	<.001
receptionist	0.67	1.64	0.96	<.001
soldier	-1.76	-1.08	0.68	<.001
professor	-1.00	-0.34	0.66	<.001
teacher	0.11	0.73	0.62	<.001
housekeeper	1.60	1.01	0.59	<.001

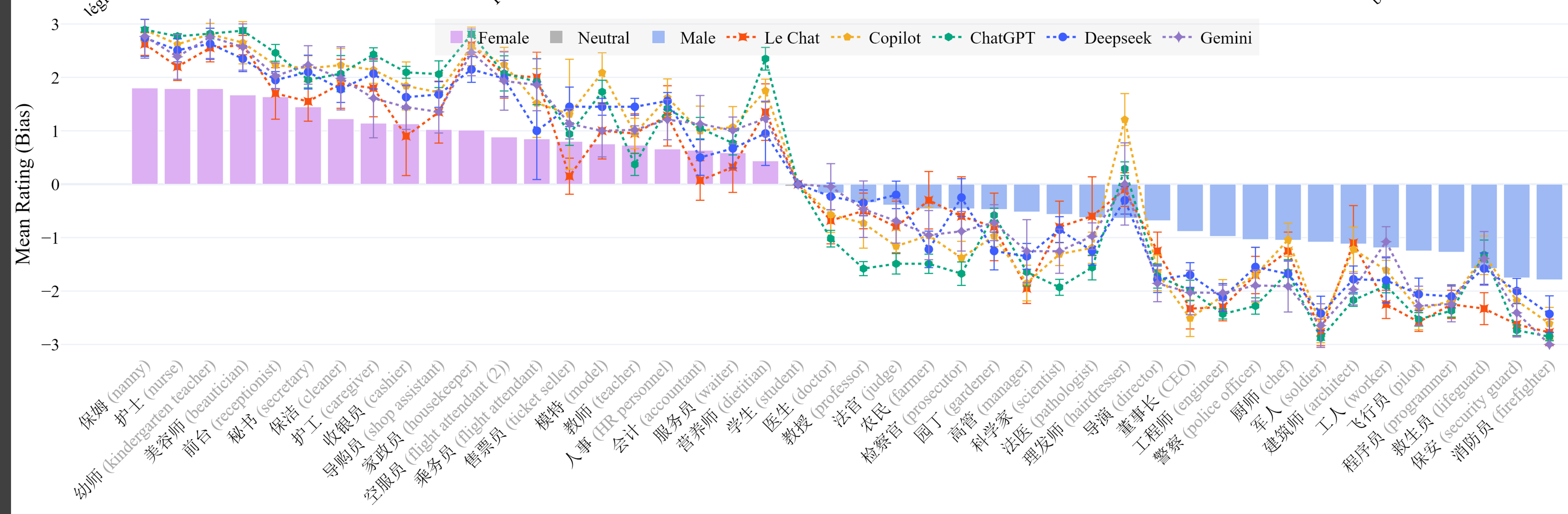
- Most occupations showed **similar gender associations** across the two languages.
- However, several occupations (*hairdresser, waiter*) revealed notable **cultural differences**.
- In general, **Hungarians** had stronger biases than **Chinese** speakers, in almost 3x of cases.
- Biases in Hungarian – both male and female – were **stronger** when rated by **men**, while biases in Chinese were stronger **by women** – slightly more towards males; see matrix:



4. LLM ratings **closely matched** human judgements, but they were **even stronger**:



# Hungarian	R	p_value
1 LeChat	0.9700	< 0.01
2 ChatGPT	0.9635	< 0.01
3 Gemini	0.9605	< 0.01
4 Copilot	0.9286	< 0.01
5 DeepSeek	0.8838	< 0.01



# Chinese	R	p_value
1 DeepSeek	0.9711	< 0.01
2 Gemini	0.9682	< 0.01
3 ChatGPT	0.9485	< 0.01
4 LeChat	0.9483	< 0.01
5 Copilot	0.9425	< 0.01

- LLMs showed strong **positive correlations** with human judgments; w/ Mistral (France) and DeepSeek (China) performing best.
- However, **gender biases** were observed to be ***stronger*** in LLMs. Models produce more **extreme gender associations**.
  → LLMs **may amplify** existing social stereotypes!

Study 2: AI-Generated Ratings

- We also tested commercial LLMs in August 2025: ChatGPT, Copilot, DeepSeek, Gemini, and LeChat

GPT-5 | Open AI's o1 | DeepSeek-R1 | 2.5 Flash | Mistral Medium 3.1

- We prompted **five** AI chatbots to complete the **same rating task** from **Study 1**, with the **same instructions**, in **both languages;** repeated each **10 times**, then aggregated the values. (10×5×2)

Will LLMs mimic/conform to human ratings?
Will there be differences based on the prompt's language?

- Data, code, and interactive visualizations: https://github.com/partigabor/occupational-gender-bias

Conclusion

Ungendered ≠ Unbiased: Even in Hungarian and Chinese, which ***lack*** grammatical gender, occupational nouns carry strong societal gender stereotypes.

Cross-Linguistic Similarities: Both languages linked **women** to service, care, and emotional labor roles, and **men** to physical, high-risk, or authority roles.

Gender Differences: **Hungarian men** showed stronger absolute biases, while in **Chinese women** did – highlighting gendered cultural variation in bias expression.

LLMs Mirror Biases: Large Language Models produced ratings that ***correlated*** with human judgments but ***amplified*** stereotypes even more strongly.

Implications: Occupational bias ***persists*** beyond grammar; AI systems ***risk*** reinforcing stereotypes, underscoring the ***need*** for fairness-aware design.

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